

Delrin® 500P BK602

ACETAL RESIN

DuPont Mobility & Materials

PROSPECTOR®

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Technical Data

Product Description

Medium Viscosity Acetal Homopolymer with Improved Processing

General

Material Status	• Commercial: Active		
UL Yellow Card ¹	• E530507-104597266		
Search for UL Yellow Card	• DuPont Mobility & Materials • Delrin®		
Availability	• Africa & Middle East • Asia Pacific	• Europe • Latin America	• North America
Additive	• Mold Release		
RoHS Compliance	• Contact Manufacturer		
Multi-Point Data	• Isothermal Stress vs. Strain (ISO 11403)	• Secant Modulus vs. Strain (ISO 11403)	
Part Marking Code (ISO 11469)	• >POM<		
Resin ID (ISO 1043)	• POM		
ISO Designation	• ISO 29988-POM-H,,MACR,4-2		

Physical	Nominal Value Unit	Test Method
Density	1.42 g/cm ³	ISO 1183
Melt Mass-Flow Rate (MFR) (190°C/2.16 kg)	15 g/10 min	ISO 1133
Melt Volume-Flow Rate (MVR) (190°C/2.16 kg)	13 cm ³ /10min	ISO 1133
Molding Shrinkage		ISO 294-4
Across Flow	1.8 %	
Flow	2.0 %	
Water Absorption		ISO 62
Saturation, 23°C, 2.00 mm	1.3 %	
Equilibrium, 23°C, 2.00 mm, 50% RH	0.40 %	

Mechanical	Nominal Value Unit	Test Method
Tensile Modulus	3100 MPa	ISO 527-1
Tensile Stress (Yield)	71.0 MPa	ISO 527-2
Tensile Strain (Yield)	14 %	ISO 527-2
Nominal Tensile Strain at Break	25 %	ISO 527-2
Flexural Modulus	3000 MPa	ISO 178
Poisson's Ratio	0.37	

Impact	Nominal Value Unit	Test Method
Charpy Notched Impact Strength		ISO 179/1eA
-30°C	7.0 kJ/m ²	
23°C	8.0 kJ/m ²	
Charpy Unnotched Impact Strength		ISO 179/1eU
-30°C	160 kJ/m ²	
23°C	200 kJ/m ²	
Notched Izod Impact Strength		ISO 180/1A
-40°C	8.0 kJ/m ²	
23°C	8.0 kJ/m ²	



Thermal	Nominal Value Unit	Test Method
Deflection Temperature Under Load		
0.45 MPa, Unannealed	162 °C	ISO 75-2/B
1.8 MPa, Unannealed	96.0 °C	ISO 75-2/A
Melting Temperature ³	178 °C	ISO 11357-3
CLTE		ISO 11359-2
Flow	1.0E-4 cm/cm/°C	
Flow : -40 to 23°C	9.0E-5 cm/cm/°C	
Transverse	1.0E-4 cm/cm/°C	
Transverse : -40 to 23°C	9.0E-5 cm/cm/°C	
Annealing Temperature	160 °C	
Annealing Time - Optional	30.0 min/mm	
Electrical	Nominal Value Unit	Test Method
Comparative Tracking Index	600 V	IEC 60112
Flammability	Nominal Value Unit	Test Method
Burning Rate ⁴ (1.00 mm)	24 mm/min	ISO 3795
Flame Rating		UL 94
0.8 mm	HB	IEC 60695-11-10, -20
1.5 mm	HB	
FMVSS Flammability	B	FMVSS 302
Additional Information	Nominal Value Unit	Test Method
Emission	< 8 ppm	VDA 275
Fogging		ISO 6452
F-value (refraction)	90 %	
G-value (condensate)	0.35 mg	
Injection	Nominal Value Unit	
Drying Temperature	80 °C	
Drying Time - Desiccant Dryer	2.0 to 4.0 hr	
Suggested Max Moisture	< 0.20 %	
Processing (Melt) Temp	210 to 220 °C	
Melt Temperature, Optimum	215 °C	
Mold Temperature	80 to 100 °C	
Mold Temperature, Optimum	90 °C	
Holding Pressure	80.0 to 100 MPa	
Drying Recommended	yes	
Hold Pressure Time	8.00 s/mm	

Notes

¹ A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.

² Typical properties: these are not to be construed as specifications.

³ 10°C/min

⁴ FMVSS 302

